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Project Technical Report – Web Technology

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Project Title: SafeReport — A Multi-Page Web Reporting Platform for Vulnerable Groups

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List of Abbreviations

- UI — User Interface
- UX — User Experience
- ER / ERD — Entity Relationship / Entity Relationship Diagram
- HTML — HyperText Markup Language
- CSS — Cascading Style Sheets
- JS — JavaScript
- ICT — Information and Communication Technology

Chapter 1. Introduction

1.1. Historical background of the case study

Reporting incidents such as abuse, neglect, or gender-based violence has traditionally relied on in-person visits to police stations, hospitals, or social welfare offices in Rwanda. For vulnerable groups — including survivors of abuse, refugees, and people with disabilities — this in-person requirement can itself be a barrier, whether due to fear, stigma, distance, or lack of trust in the reporting channel. In recent years, Rwanda has invested heavily in e-government services, most visibly through the Irembo platform, which has demonstrated that citizens are willing to use digital channels for sensitive administrative processes when the platform is simple, trustworthy, and available in local languages. Safereport was conceived against this backdrop, as a coursework project exploring how a similar digital-first, trust-oriented approach could be applied to incident reporting for vulnerable groups.

1.2. Problem statement

Vulnerable individuals in Rwanda who experience abuse, neglect, or related harms often lack a single, accessible, and confidential channel through which to report their situation and be connected to the appropriate support service (health, legal, psychological, shelter, and so on). Existing channels are fragmented across different offices and hotlines, are not always available in the reporter's preferred language, and typically leave the reporter with no way to track what happens to their report after it is submitted. Safereport addresses this gap by providing a single web-based entry point for reporting, automatic routing to relevant support services by category, and a reference code the reporter can use to follow up.

1.3. Objectives of your project

General objective: to design and implement a multi-page web platform that allows vulnerable individuals to report incidents safely and be connected to appropriate support services.

Specific objectives:

- To design a report form that supports both anonymous and named submissions.
- To automatically match a submitted report's category to relevant support services and display them to the reporter.
- To generate a unique reference code per report so a reporter can track the status of their case.
- To provide a basic administrative view so reports can be reviewed and their status updated.
- To support multiple languages (English, Kinyarwanda) so the platform is accessible to a wider range of users.

1.4. Expected Outcomes

By the end of the project, Safereport is expected to provide: a working multi-page site (welcome, home, about, report form, confirmation, and admin pages); client-side form validation; a functioning case-tracking mechanism using a generated reference code; a password-protected administrative dashboard for reviewing and updating report status; and a partially implemented multilingual interface, with the remaining pages and the ATTACHMENT feature identified as follow-up work beyond the current submission.

1.5. Project Rationale (Importance to the community)

A confidential, accessible reporting channel can lower the barrier for vulnerable individuals to seek help, particularly in cases where visiting a physical office is unsafe, stigmatizing, or impractical. By routing reports to the correct category of support service and giving reporters a way to track their case, Safereport aims to reduce the chance that a report is lost, ignored, or misdirected, and to make the process feel more dignified and less intimidating for the people who use it.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1 Intended users of the project

- Members of the public, particularly vulnerable individuals (survivors of abuse, refugees, people with disabilities, and others), who need to report an incident and find relevant support.
- System administrators / support staff, who review incoming reports and update their status as they are handled.

2.1.2 Intended system partners

Safereport is designed to eventually route reports to real-world partner organizations corresponding to each report category, including:

- Health services offices
- Legal aid centers
- Trauma and abuse support centers
- Gender and advocacy offices (GBV)
- Youth counseling / mental health and psychological support lines
- Child protection units
- Anti-trafficking hotlines
- Emergency shelter networks
- Refugee support offices
- Disability rights centers
- Financial aid and welfare offices

In the current implementation these are represented as placeholder contact cards matched by category; connecting them to real partner organizations is future work.

2.2. System Design

High-Level Architecture

SafeReport is currently implemented as a client-side application: a set of HTML pages sharing one stylesheet and one JavaScript file (`safereport.js`), with report data persisted in the browser's `localStorage` rather than a server-side database. This keeps the project achievable within the coursework's scope while still demonstrating the full reporting and case-management workflow. Figure 1 shows this architecture, including the planned future migration to a shared backend and database.

Figure: SafeReport High-Level Architecture

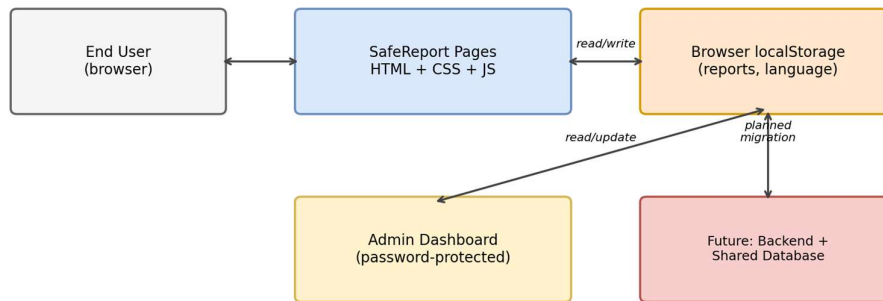
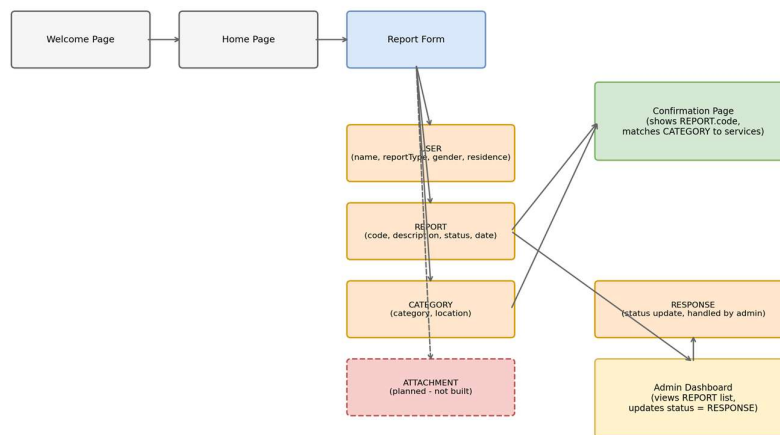


Figure 1: SafeReport High-Level Architecture

UI design (sketch of the UI module) and Flowcharts

Figure 2 shows how the platform's pages connect to one another and to the underlying data entities: a user moves from the Welcome page through the Home page to the Report Form, which creates a REPORT (linked to a USER and a CATEGORY); the Confirmation page then displays the generated reference code and matches the report's category to support services; and the Admin Dashboard separately reads all REPORT records and writes status updates, modeled as the RESPONSE entity.

SafeReport - Page Flow linked to ER Entities



Solid box = ER entity | Dashed box/edge = planned, not yet built
Green = existing page | Yellow = admin-only page

Figure 2: SafeReport Page Flow linked to ER Entities

Database design with ERD and relationships

Although the current implementation stores data in localStorage rather than a relational database, the system was designed around the entity-relationship model shown in Figure 3, so that migrating to a real database (identified as future work in Chapter 4) follows a clear schema. A USER can submit many REPORTs; each REPORT belongs to one CATEGORY; each REPORT may have many ATTACHMENTs (planned, not yet implemented) and many RESPONSE entries recording status changes made by an administrator.

Figure: SafeReport Entity Relationship Diagram

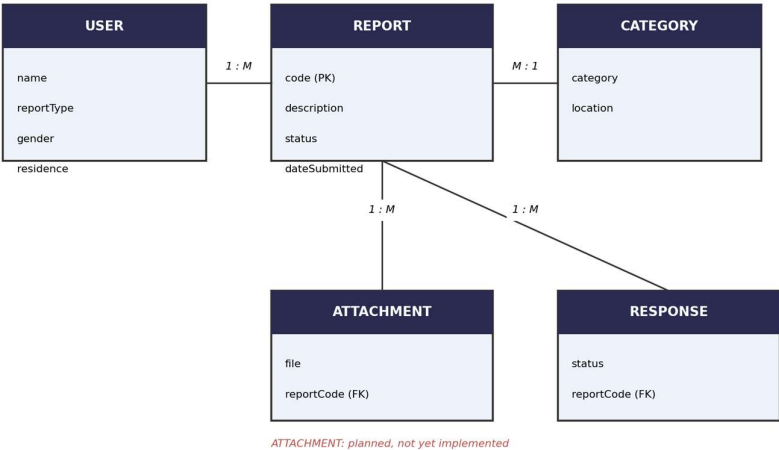


Figure 3: SafeReport Entity Relationship Diagram

Chapter 3. Implementation

3.1. Introduction of the section

This section presents the implemented pages of SafeReport with screenshots and a short discussion of each. Insert a screenshot of the live page (or a local browser screenshot) in each boxed placeholder below before submission.

3.2.1 Welcome Page

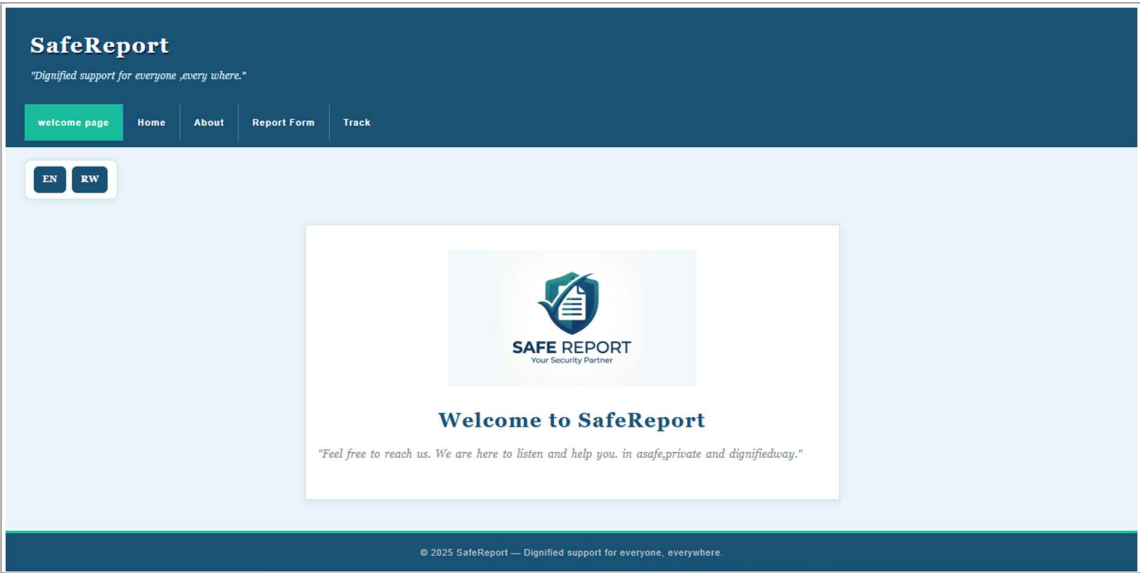


Figure 4: Welcome page

The welcome page is the first entry point of SafeReport. It introduces the platform's purpose, "Dignified support for everyone, everywhere," and directs the visitor toward the Home page. The design uses a dark navy and teal palette intended to feel calm and trustworthy for people who may be visiting the site in a moment of distress.

3.2.2 Home Page

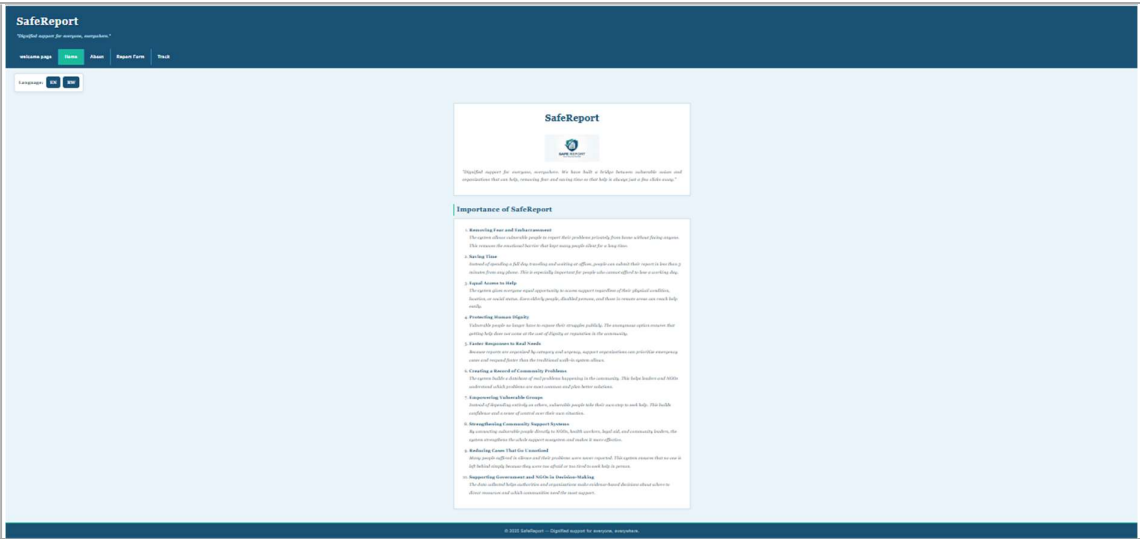


Figure 5: Home page

The Home page orients the user to the platform's services and provides navigation to the About page, the Report Form, and support information. It is designed to be simple and low-friction, since users accessing SafeReport may have limited time, privacy, or digital literacy.

3.2.3 About Page

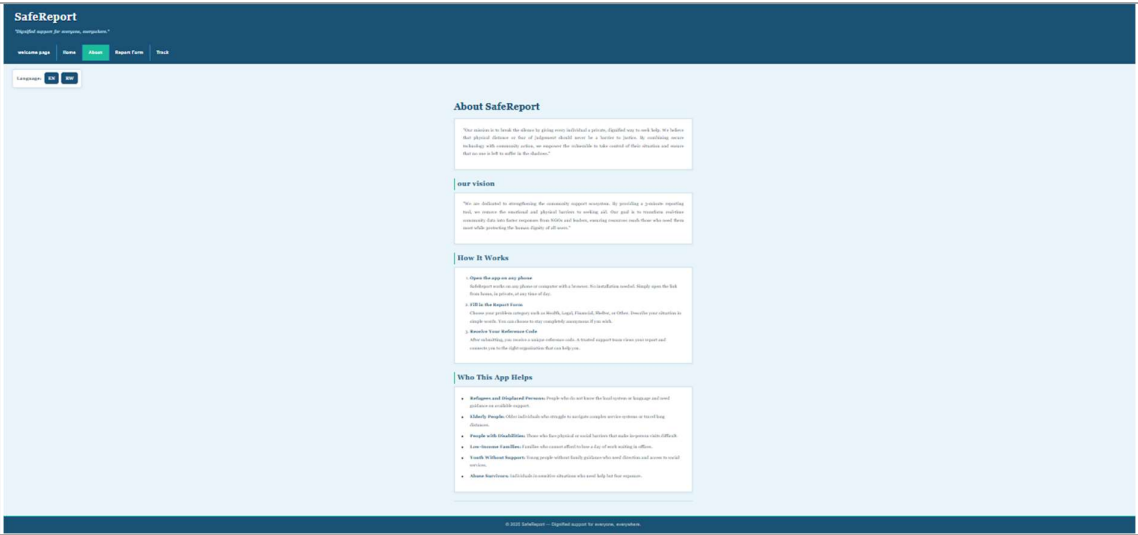


Figure 6: About page

The About page explains SafeReport's mission, confidentiality commitments, and the categories of support available. It is intended to build trust before a user commits to filling in the report form.

3.2.4 Report Form Page

Figure 7: Report form with all fields visible

The report form allows a user to submit a report either anonymously or with their name, select their district of residence, choose a category (e.g. Health, Legal Rights, Child Protection), and describe their situation. Client-side JavaScript validation (`validate_form`) ensures the description field is filled in and, if provided, the email is well-

2formed. On submission, a unique reference code is generated (generateCaseCode) and the report is saved to the browser's localStorage (saveReport) before the user is redirected to the confirmation page.

3.2.5 Confirmation Page

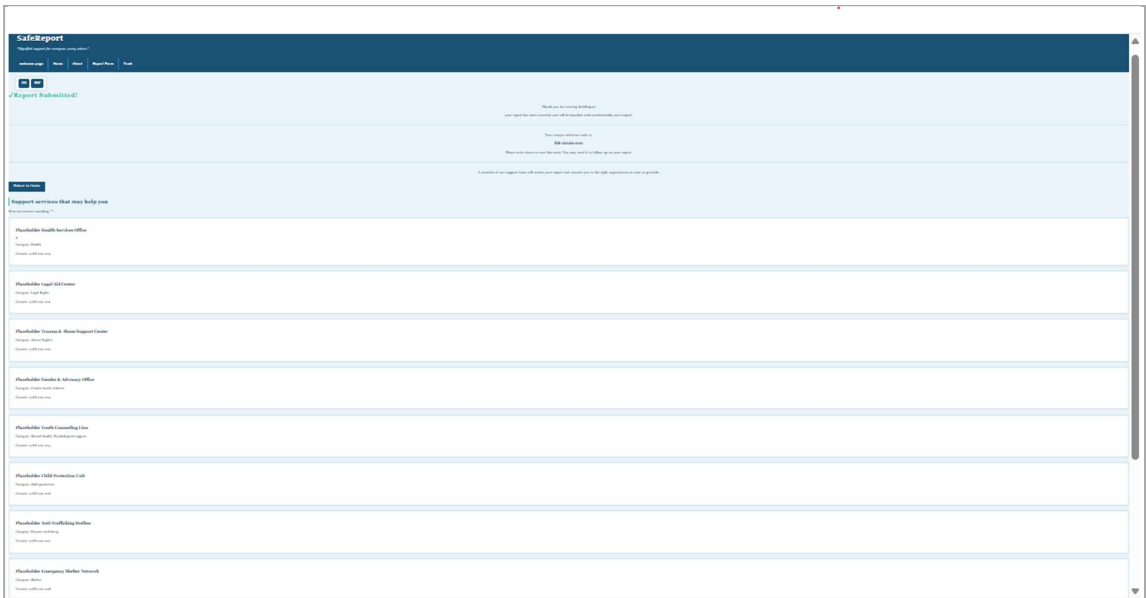


Figure 8: Confirmation page showing the reference code and matched services

After submission, the confirmation page displays the generated reference code prominently and instructs the user to save it for follow-up. It also calls showMatchingServices to filter and display the support-service cards relevant to the category the user selected, so the user immediately sees where to seek further help.

3.2.6 Admin Dashboard



Figure 9: Admin dashboard listing submitted reports

The admin dashboard (safereport_admin.html) is a password-protected page (checkAdminAccess) that lists every report currently stored in localStorage, including its reference code, category, district, description, status, and

submission date. An admin can update a report's status (Submitted, In Review, Resolved) via `updateStatus`, which is the system's basic case-management workflow.

3.2.7 Language Toggle



Figure 10: Report form shown in English, Kinyarwanda, and French

A language switcher (EN / RW) is present in the navigation bar on the pages it has been implemented on so far. Selecting a language calls `setLanguage`, which rewrites every element tagged with `data-i18n` using a JavaScript translation dictionary, and persists the choice in `localStorage` via `loadLanguage` so it is remembered on the next visit.

Chapter 4. Conclusion and Recommendation

At the time of this report, the core reporting workflow, case-tracking mechanism, and administrative dashboard are fully implemented and functional, representing the majority of the project's specific objectives. The multilingual interface is partially implemented — the navigation bar and report form support English, Kinyarwanda, and French, while the remaining pages (Home, About, Welcome) and the category dropdown options still need translation tags applied. The attachment-upload feature identified in the system design has not yet been implemented. Overall, approximately 70–80% of the stated objectives have been achieved as of this submission, with the remainder identified as near-term follow-up work.

Recommendations

To the lecturer, regarding technicalities: the coursework's requirement to use specific JavaScript conventions (var, named functions, inline event handlers) was followed throughout, which constrained some modern JavaScript patterns that would otherwise simplify the code (for example, template literals or arrow functions); flexibility on this point in future modules could allow more idiomatic code.

To the school, regarding time constraints: implementing a genuinely multilingual, accessible reporting platform — including real backend storage — is a substantial undertaking; more time or a staged submission schedule would allow features such as attachments and shared (non-localStorage) data to be completed to a higher standard.

To potential partners (the organizations SafeReport would route reports to): the placeholder contact information in this version would need to be replaced with verified, up-to-date contacts before any real deployment, and each partner organization would need to agree on a process for receiving and acting on routed reports.

To users: because the current version stores data only in the browser used to submit the report, users should be advised to submit and check on the same device and browser until a shared backend is implemented.

To beneficiaries: as the platform matures, training and outreach will be needed so that the intended beneficiaries — vulnerable individuals who may have limited digital literacy — are aware SafeReport exists and understand that their reports are handled confidentially.

Appendix

Link to the GitHub repository of the deployed project (and login credentials, if applicable):

[<https://uannechristella-lgtm.github.io/updated-safereport-project/>]

Password for dashboard is safereport2026